Edge Detection

Shree K. Nayar

Columbia University

Topic: Edge Detection, Module: Features

First Principles of Computer Vision

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Edge Detection

Convert a 2D image into a set of points where image intensity changes rapidly.

Topics:

(1) What is an Edge?

(2) Edge Detection Using Gradients

(3) Edge Detection Using Laplacian

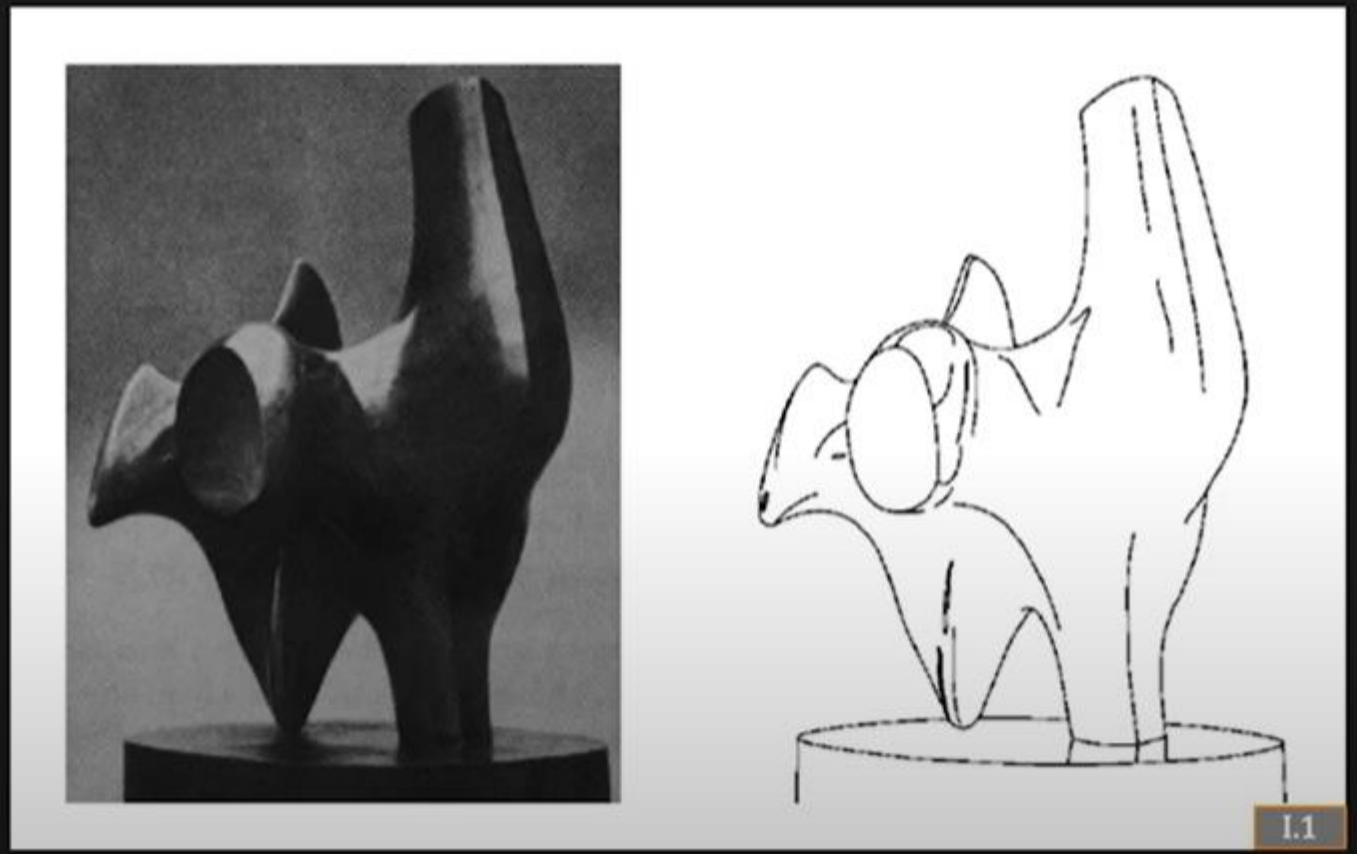
(4) Canny Edge Detector

(5) Corner Detection

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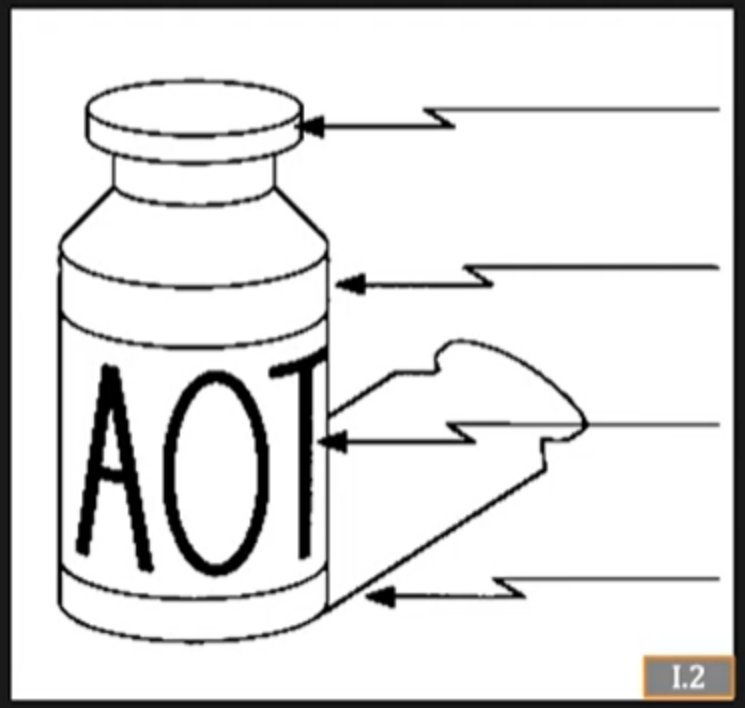
What is an Edge?

Rapid change in image intensity within small region



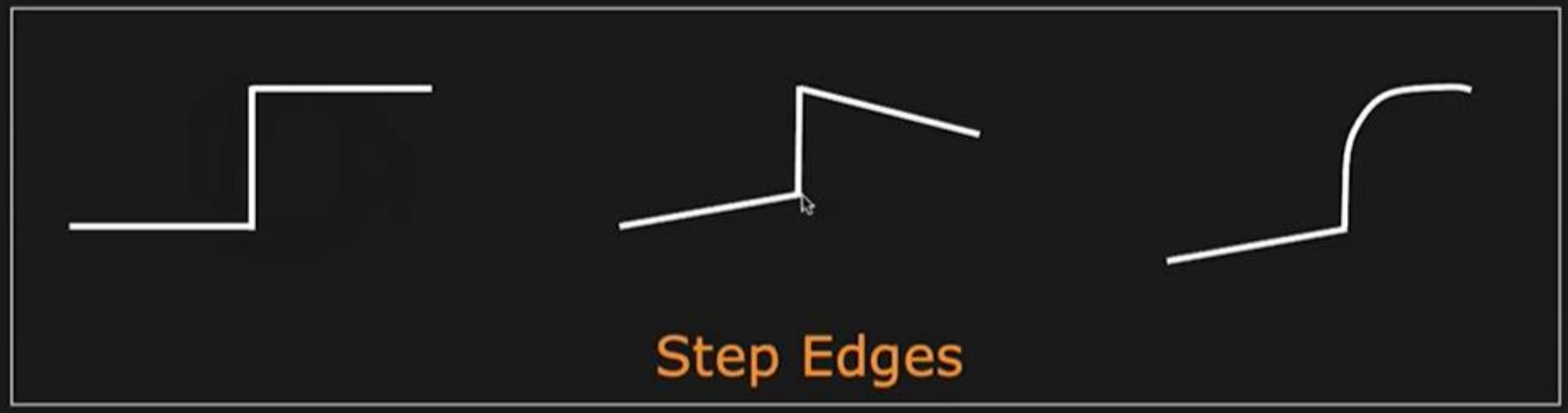
Causes of Edges

Rapid changes in image intensity are caused by various physical phenomena.



- Surface Normal Discontinuity
- Depth Discontinuity
- Surface Reflectance Discontinuity
- Illumination Discontinuity

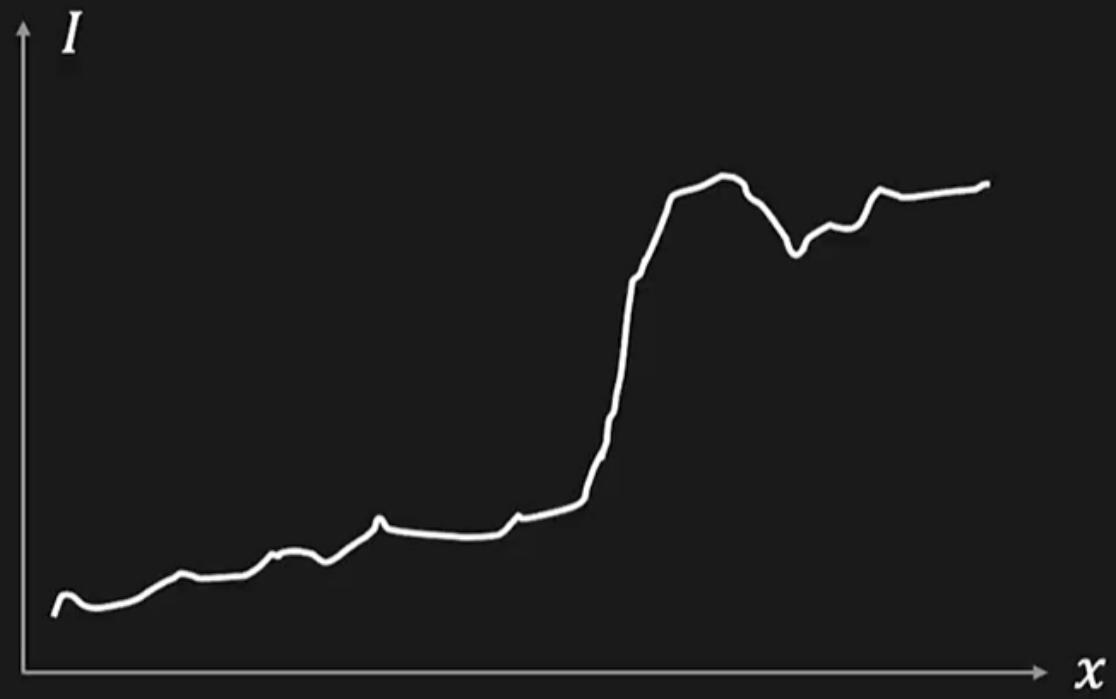
Types of Edges







Real Edges



Problems: Noisy Images and Discrete Images

Edge Detector

We want an Edge Operator that produces:

- Edge Position
- Edge Magnitude (Strength)
- Edge Orientation (Direction)

Performance Requirements:

- High Detection Rate
- Good Localization
- Low Noise Sensitivity

Edge Detection Using Gradients

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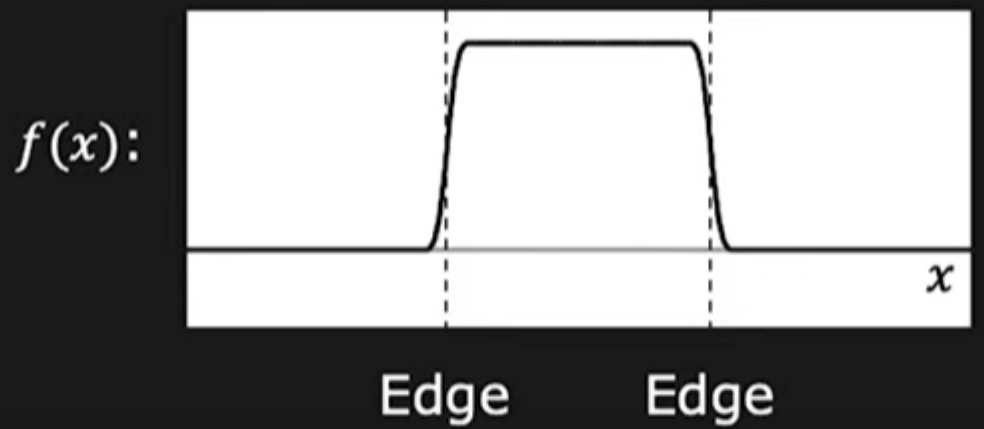
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Topic: Edge Detection, Module: Features

First Principles of Computer Vision

1D Edge Detection

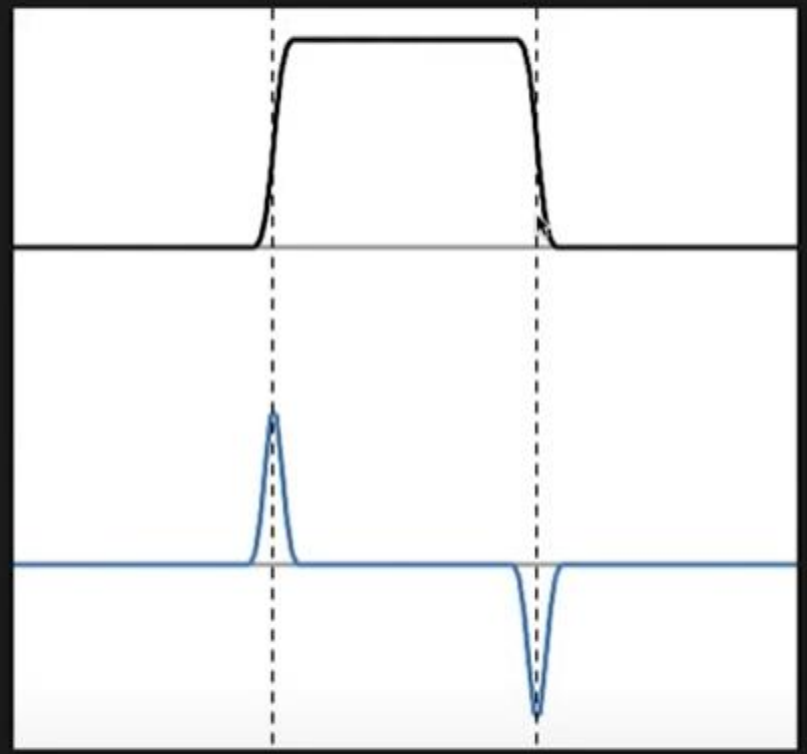
Edge is a rapid change in image intensity in a small region.



Basic Calculus: Derivative of a continuous function represents the amount of change in the function.

Edge Detection Using 1st Derivative

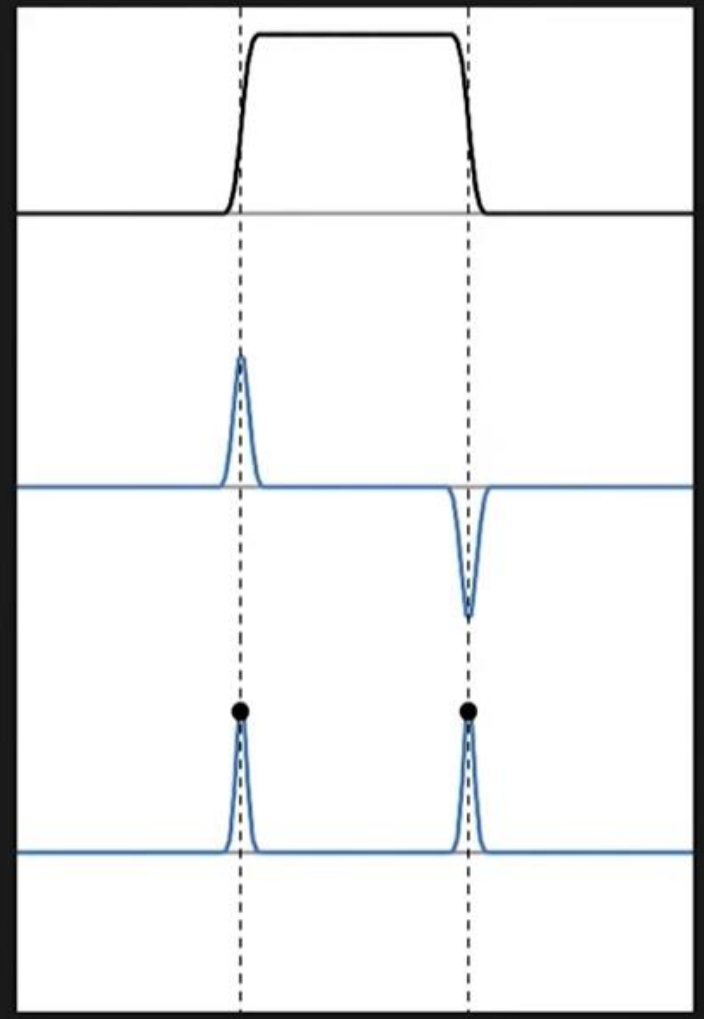
$f(x)$



First Derivative: $\frac{\partial f}{\partial x}$

Local Extrema
Indicate Edges

Edge Detection Using 1st Derivative



First Derivative: $\frac{\partial f}{\partial x}$

Local Extrema
Indicate Edges

First Derivative
Absolute Value: $\left|\frac{\partial f}{\partial x}\right|$

Local Maxima
Indicate Edges

Provides Both Location and Strength of an Edge

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2D Edge Detection

$I(x, y)$:



Edge

Edge

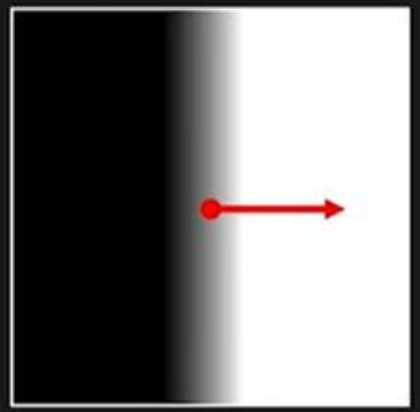
Basic Calculus: Partial Derivatives of a 2D continuous functi
represents the amount of change along each dimension.

Gradient (∇)

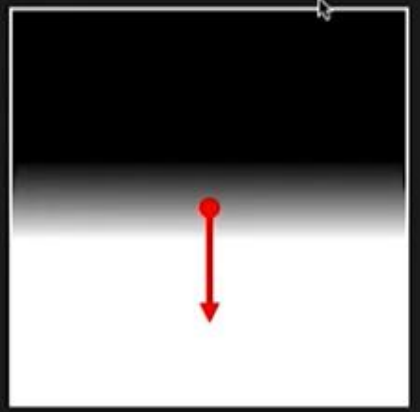
Gradient (Partial Derivatives) represents the direction of most rapid change in intensity

$$\nabla I = \left[\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}\right]$$

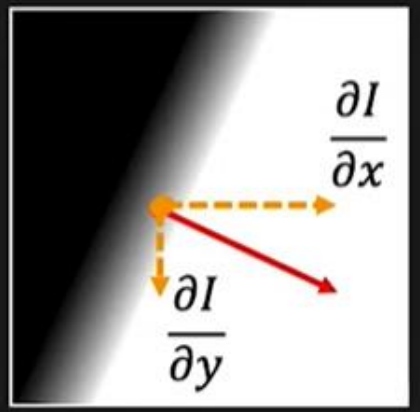
Pronounced as "Del I"



$$\nabla I = \left[\frac{\partial I}{\partial x}, 0\right]$$



$$\nabla I = \left[0, \frac{\partial I}{\partial y}\right]$$

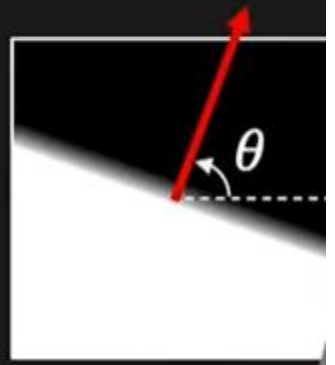


$$\nabla I = \left[\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}\right]$$

Gradient (∇) as Edge Detector

Gradient Magnitude $S = \|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2}$

Gradient Orientation $\theta = \tan^{-1}\left(\frac{\partial I}{\partial y} / \frac{\partial I}{\partial x}\right)$

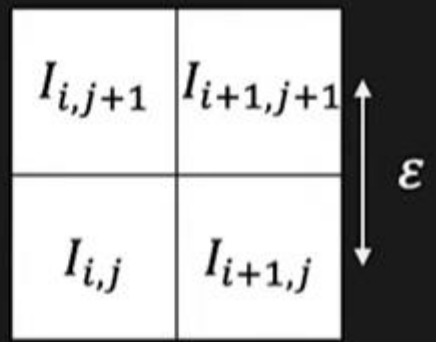


Discrete Gradient (∇) Operator

Finite difference approximations:

$$\frac{\partial I}{\partial x} \approx \frac{1}{2\varepsilon}\left(\left(I_{i+1,j+1} - I_{i,j+1}\right) + \left(I_{i+1,j} - I_{i,j}\right)\right)$$

$$\frac{\partial I}{\partial y} \approx \frac{1}{2\varepsilon}\left(\left(I_{i+1,j+1} - I_{i+1,j}\right) + \left(I_{i,j+1} - I_{i,j}\right)\right)$$

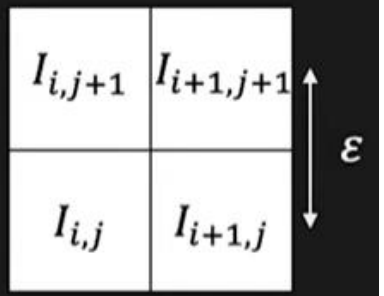


Discrete Gradient (∇) Operator

Finite difference approximations:

$$\frac{\partial I}{\partial x} \approx \frac{1}{2\varepsilon}\left(\left(I_{i+1,j+1} - I_{i,j+1}\right) + \left(I_{i+1,j} - I_{i,j}\right)\right)$$

$$\frac{\partial I}{\partial y} \approx \frac{1}{2\varepsilon}\left(\left(I_{i+1,j+1} - I_{i+1,j}\right) + \left(I_{i,j+1} - I_{i,j}\right)\right)$$



Can be implemented as Convolution!

$$\frac{\partial}{\partial x} \approx \frac{1}{2\varepsilon}\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \qquad \frac{\partial}{\partial y} \approx \frac{1}{2\varepsilon}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

Note: Convolution flips have been applied

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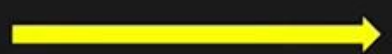
Comparing Gradient (∇) Operators

Gradient	Roberts	Prewitt	Sobel (3x3)	Sobel (5x5)
$\frac{\partial I}{\partial x}$	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & -2 & 0 & 2 & 1 \\ -2 & -3 & 0 & 3 & 2 \\ -3 & -5 & 0 & 5 & 3 \\ -2 & -3 & 0 & 3 & 2 \\ -1 & -2 & 0 & 2 & 1 \end{bmatrix}$
$\frac{\partial I}{\partial y}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 3 & 2 & 1 \\ 2 & 3 & 5 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ -2 & -3 & -5 & -3 & -2 \\ -1 & -2 & -3 & -2 & -1 \end{bmatrix}$

Comparing Gradient (∇) Operators

Gradient	Roberts	Prewitt	Sobel (3x3)	Sobel (5x5)
$\frac{\partial I}{\partial x}$	0 1 -1 0	-1 0 1 -1 0 1 -1 0 1	-1 0 1 -2 0 2 -1 0 1	-1 -2 0 2 1 -2 -3 0 3 2 -3 -5 0 5 3 -2 -3 0 3 2 -1 -2 0 2 1
$\frac{\partial I}{\partial y}$	1 0 0 -1	1 1 1 0 0 0 -1 -1 -1	1 2 1 0 0 0 -1 -2 -1	1 2 3 2 1 2 3 5 3 2 0 0 0 0 0 -2 -3 -5 -3 -2 -1 -2 -3 -2 -1

Good Localization
Noise Sensitive
Poor Detection



Poor Localization
Less Noise Sensitive
Good Detection

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Gradient (∇) Using Sobel Filter



Image (I)



$\partial I/\partial x$



$\partial I/\partial y$



Gradient Magnitude

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Edge Thresholding

Standard: (Single Threshold T)

$\|\nabla I(x,y)\| < T$ Definitely Not an Edge

$\|\nabla I(x,y)\| \geq T$ Definitely an Edge

Edge Thresholding

Standard: (Single Threshold T)

$\|\nabla I(x,y)\| < T$ Definitely Not an Edge

$\|\nabla I(x,y)\| \geq T$ Definitely an Edge

Hysteresis Based: (Two Thresholds $T_0 < T_1$)

$\|\nabla I(x,y)\| < T_0$ Definitely Not an Edge

$\|\nabla I(x,y)\| \geq T_1$ Definitely an Edge

$T_0 \leq \|\nabla I(x,y)\| < T_1$ Is an Edge if a Neighboring Pixel is Definitely an Edge

Sobel Edge Detector



Image (I)



$\partial I / \partial x$



$\partial I / \partial y$



Gradient Magnitude



Thresholded Edge

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